

Symmetry Protected Quantum States

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Introduction

This is a concise report of my research project in quantum computation and information as a summer student in Professor M. Szymanska's group at UCL, focusing on intrinsically symmetry-protected Hamiltonian models for quantum information and computation, which was largely based on the ideas of Dr. Yi et al. The author reproduced some known results but also extended the investigation to higher-spin ($N > 4$) systems and open quantum system evolution. The first section mainly focuses on the analysis of the decoherence sensitivity of a multi-qubit symmetry-protected Hamiltonian model, while the second half deals with master equation evolution on the given model subject to simple Pauli noises from coupling with the environment. The author is grateful for the opportunity in Szymanska's group supported by the CMMP summer bursary funding and fruitful discussions with Dr. Yi in particular.

Symmetry-Protected Spin Model

For the purpose of realizing quantum circuits, the field of quantum error correction has seen significant advances. The standard approaches tend to rely mostly on storing multiple physical qubits to represent a single logical qubit. However, this is not without issues, as scaling up the number of physical qubits can be realistically difficult and inefficient.

However, the essential idea is still valuable. For quantum error correction, the general assumption is that single-qubit errors are much more frequent than multiple-qubit ones, so the approach to store quantum information non-locally in many qubits protects the system from local perturbations.

Taking this idea further, we could use a Hamiltonian model (of at least 4 spins), which contains terms of nearest-neighbour coupling and next-nearest-neighbour coupling respectively. One can then utilize the lowest two eigenstates to form a symmetry-protected subspace, where both relaxation and dephasing errors are eliminated up to first order.

The many-body spin model of M (> 4) qubits is given by the following expression:

$$H_0 = \frac{t}{2} \sum_{m=1}^M (\sigma_m^+ \sigma_{m+1}^- + \sigma_m^- \sigma_{m+1}^+) + \frac{\lambda}{2} \sum_{m=1}^M (\sigma_m^+ \sigma_{m+2}^- + \sigma_m^- \sigma_{m+2}^+), \quad (1)$$

where t is the nearest-neighbour coupling parameter, and λ is the next-nearest-neighbour coupling parameter, and coupling scheme is illustrated in *Figure 1*. Here σ_m^+ and σ_m^- are the creation and annihilation Pauli operators respectively, acting on the m -th qubit, defined as

$$\sigma_m^\pm = \frac{\sigma_m^x \pm i\sigma_m^y}{2}, \quad (2)$$

where we also assume periodic boundary conditions such that $\sigma_{M+1} = \sigma_1$ and $\sigma_{M+2} = \sigma_2$.

We would demonstrate that for a given range of λ/t , both relaxation and dephasing caused by local Pauli noises can be canceled to first order and become negligible.

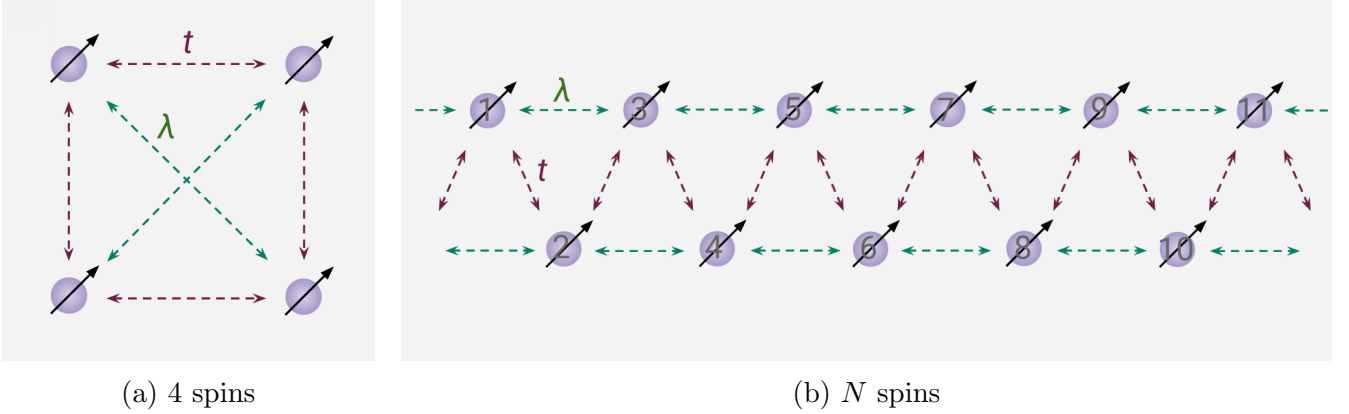


Figure 1: Hamiltonian spin model couplings [1].

To analyze why this holds, we need to check where the "symmetry" of the Hamiltonian model is really well-protected. To this end, we have to define several key symmetry operators $\mathcal{N}$ (number), $\mathcal{T}$ (translation) and $\mathcal{I}$ (inversion) as in [2], and they satisfy the following key relations:

$$\mathcal{N} = \sum_{m=1}^M \frac{1 + \sigma_m^z}{2}. \quad (3)$$

$$\mathcal{T} \sigma_m \mathcal{T}^{-1} = \sigma_{m+1}. \quad (4)$$

$$\mathcal{I} \sigma_m \mathcal{I} = \sigma_{M+1-m}. \quad (5)$$

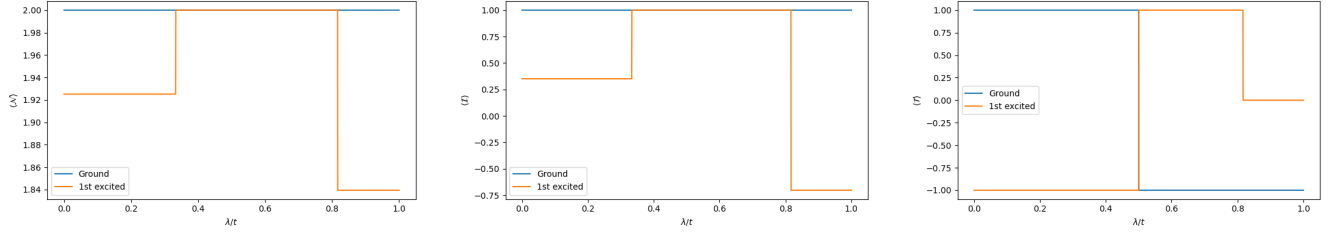
For our Hamiltonian model in (1), defining the ground eigenstate as $|0\rangle$ and the first excited state as $|1\rangle$, we expect the following eigenvalues for our above symmetry operators in *Table 1*. Doing

Table 1: Eigenvalues of symmetry operators.

Operators	$ 0\rangle$	$ 1\rangle$
$\mathcal{N}$	$M/2$	$M/2$
$\mathcal{I}$	1	1
$\mathcal{T}$	± 1	∓ 1

a numerical computation with just 4 spins ($M = 4$), one obtains the following diagrams for the eigenvalues in *Figure 2*. Comparing the results with our expected values in *Table 1*, one can expect symmetry to be preserved roughly around λ/t in the range from 0.4 to 0.8, which will be later confirmed with more numerical simulations.

Now we must connect the relationship between symmetry protection for these eigenstates and protection against perturbations or errors. It is thus necessary to define exactly the relaxation and



(a) Number symmetry $\mathcal{N}$

(b) Inversion symmetry $\mathcal{I}$

(c) Translation symmetry $\mathcal{T}$

Figure 2: Simulated eigenvalues indicating the symmetry-protected range in λ/t .

dephasing amplitudes (R and D) in our model:

$$R = \sqrt{\sum_{w=x,y,z} |\langle 1 | \sigma_m^w | 0 \rangle|^2}. \quad (6)$$

$$D = \sqrt{\sum_{w=x,y,z} |\langle 1 | \sigma_m^w | 1 \rangle - \langle 0 | \sigma_m^w | 0 \rangle|^2}. \quad (7)$$

One can show theoretically that conforming to the eigenvalues in *Table 1* help eliminate R and D , thereby meaning a robust protection against possible errors [2].

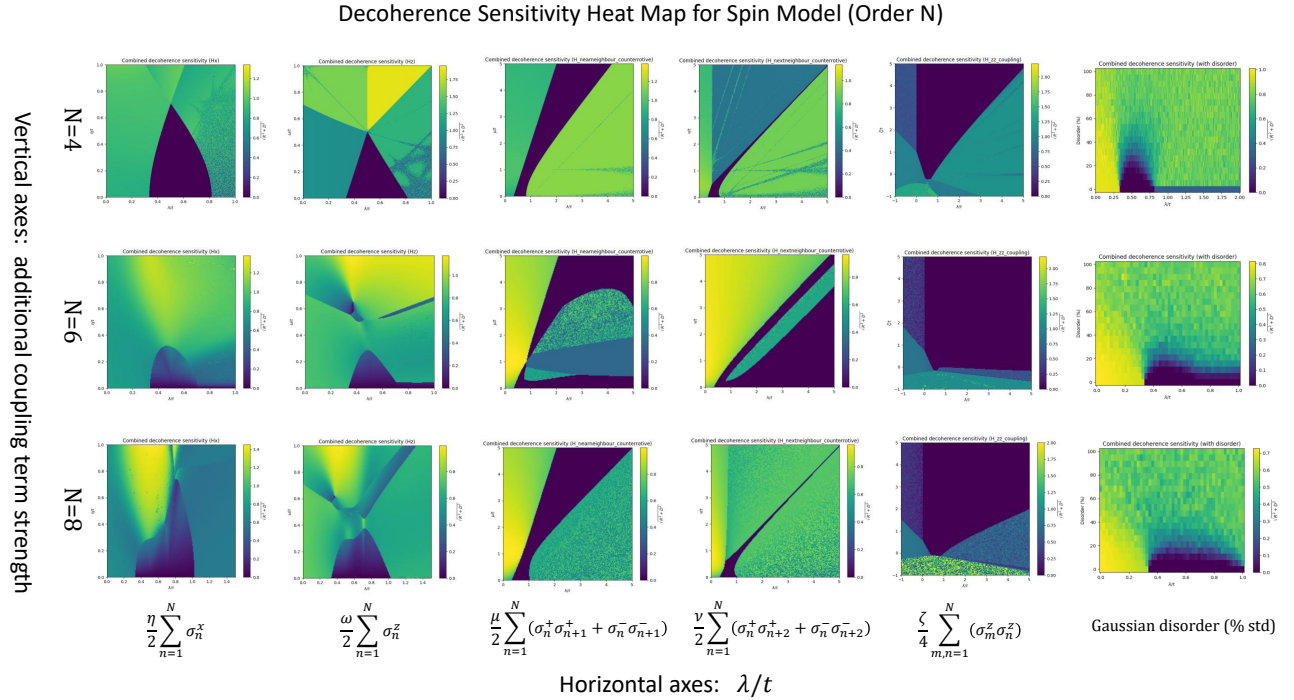


Figure 3: Decoherence sensitivity for various spins with additional terms or Gaussian noise. See *Appendix* for more clear images individually.

In realistic systems, there are often additional residue terms added to (1) which will affect the

performance of our existing model. To model how decoherence sensitivity $\sqrt{R^2 + D^2}$ behaves, one can plot a heatmap as in *Figure 3*, whereby the colour demonstrates the magnitude of decoherence sensitivity, the horizontal axis is the magnitude of λ/t , and the vertical axis is the magnitude (relative to t) of the parameter of the additional residue term considered. In the compiled figure, each row is for a specific spin, and we simulated $N = 4, 6, 8$ respectively. From left to right (except the rightmost column), the columns respond to cases of transverse field, longitudinal field, short-range counter-rotative terms, long-range counter-rotative terms and all-to-all σ^z - σ^z couplings respectively. The last column is Gaussian random noise applied to (1) with varying standard deviation (%) set by the vertical axis. Generally, one can observe a dark purple region where $\sqrt{R^2 + D^2}$ is close to zero, and one can always take this regime in which $|0\rangle$ and $|1\rangle$ are well-behaved robust logical qubits.

To further examine the case with Gaussian random noise, take just the situation where $\lambda/t = 0.6$, one can make a plot of various N (number of spins) for the decoherence sensitivity under different magnitudes of noise, as in *Figure 4*.

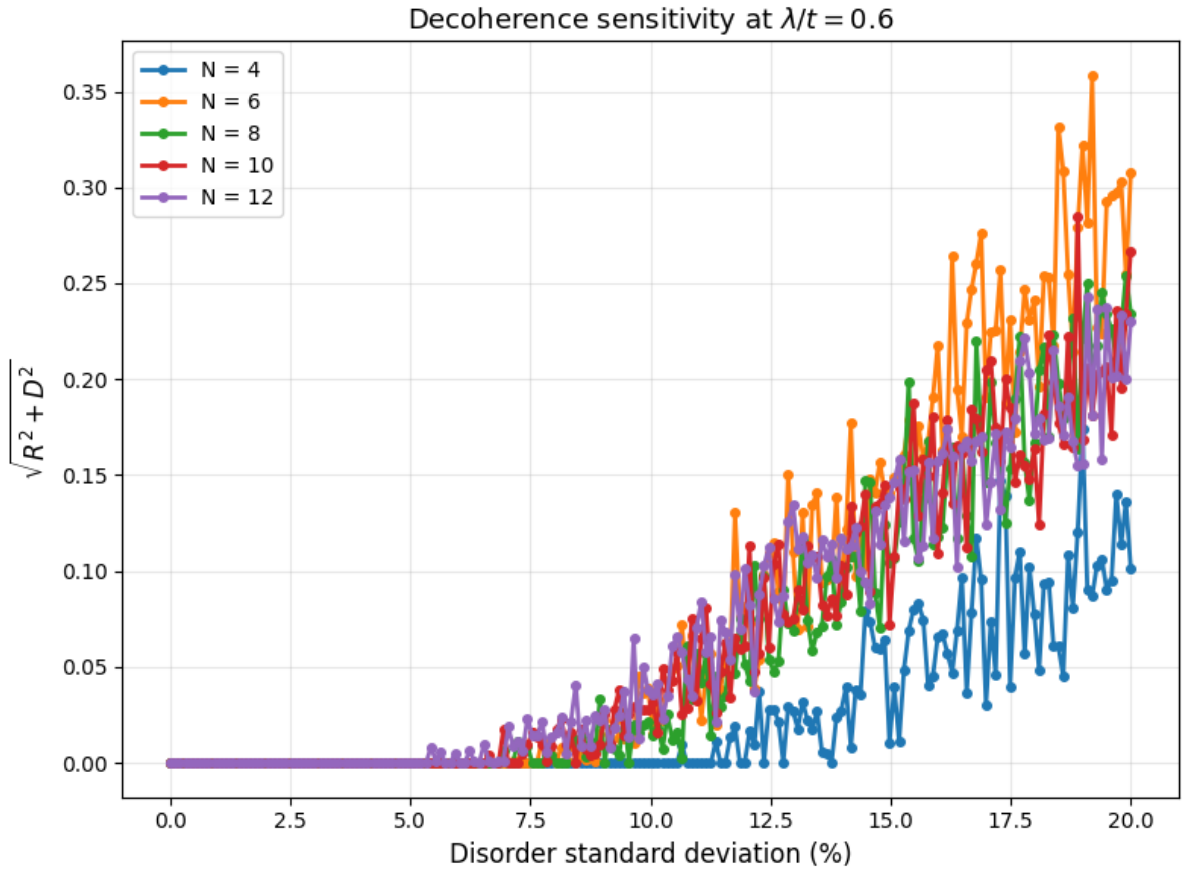


Figure 4: Effects of Gaussian noise on decoherence sensitivity, $\lambda/t = 0.6$.

It is possible to see that the $N = 4$ case behaves most optimally with the least absolute magnitude of decoherence amplitude, whilst higher-spin models do behave more or less similar in magnitude. However, if one is interested in the decoherence amplitude per spin, then it follows immediately that

higher-spin models could minimize this effectively, and storing quantum information non-locally between many qubits do help minimize the mean error.

Master Equation Evolution

Another important aspect to test our model is through the formalism of open quantum systems, where the target system couples to the environment and is subject to noise [3]. Particularly, the Franke–Gorini–Kossakowski–Sudarshan–Lindblad master equation in (8) helps establish a quantitative framework for numerical simulations.

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} \right). \quad (8)$$

In (8), ρ is the density operator of our quantum state, H is the spin Hamiltonian containing the t and λ flip-flop couplings. L_k is the collapse operator acting on the k -th qubit, determining the degree of environmental coupling to the system.

In time evolution, it is useful to define the fidelity $F(t)$ of the density operator $\rho(t)$ at time t as

$$F(t) = |\langle \psi(0) | \rho(t) | \psi(0) \rangle|, \quad (9)$$

where $|\psi(0)\rangle$ is the (pure) state at $t = 0$ initially.

Now we consider the limiting case where $L_k = 0$ for all k , then (8) reduces to the von Neumann equation $d\rho/dt = -i[H, \rho]$, and we need to tell two cases apart: First, if the state $|\psi(0)\rangle$ is an eigenstate of H , then $|\psi(0)\rangle \rightarrow e^{-iE_0 t} |\psi(0)\rangle$ with only a phase shift from unitary evolution, and $F(t) = 1$ as the global phases cancel out exactly in the calculation. Second, if the state is general (i.e. being in a superposition state of $|0\rangle$ and $|1\rangle$ or more), then $|\psi(0)\rangle = \alpha |0\rangle + \beta |1\rangle \rightarrow \alpha e^{-iE_0 t} |0\rangle + \beta e^{-iE_1 t} |1\rangle$, and with some algebra one can show that $F(t) = 1 - 4|\alpha|^2|\beta|^2 \sin^2((E_1 - E_0)t/2)$, which is a trigonometric function in harmonic oscillation. This oscillatory pattern can be generalized to a superposition of more than 2 eigenstates.

Retrieving back the case $L_k \neq 0$, avoiding complicated maths, one would qualitatively expect an envelope of exponential decay imposed on the fidelity of the density operator. For eigenstates this is just a simple exponential decay (as imposed only at a constant $F = 1$), but for non-eigenstates this would be an exponentially decaying oscillation.

To illustrate what happens to our symmetry-protected eigenstates (i.e. the ground state $|0\rangle$ here) from the original Hamiltonian model in (1), we could consider plots for various versions of noise in the form of different Lindblad collapse operators. For simplicity, one can calculate the evolution of fidelity for cases where $L_k = \sqrt{\gamma}\sigma_k^x$ in *Figure 5(a)*, $\sqrt{\gamma}\sigma_k^z$ in *Figure 5(b)* and $\sqrt{\gamma/2}(\sigma_k^x + \sigma_k^z)$ in *Figure 5(c)* respectively, corresponding to single σ^x , σ^z noises or both superposed, with γ being the parameter controlling the magnitude of environmental noise. For the states and

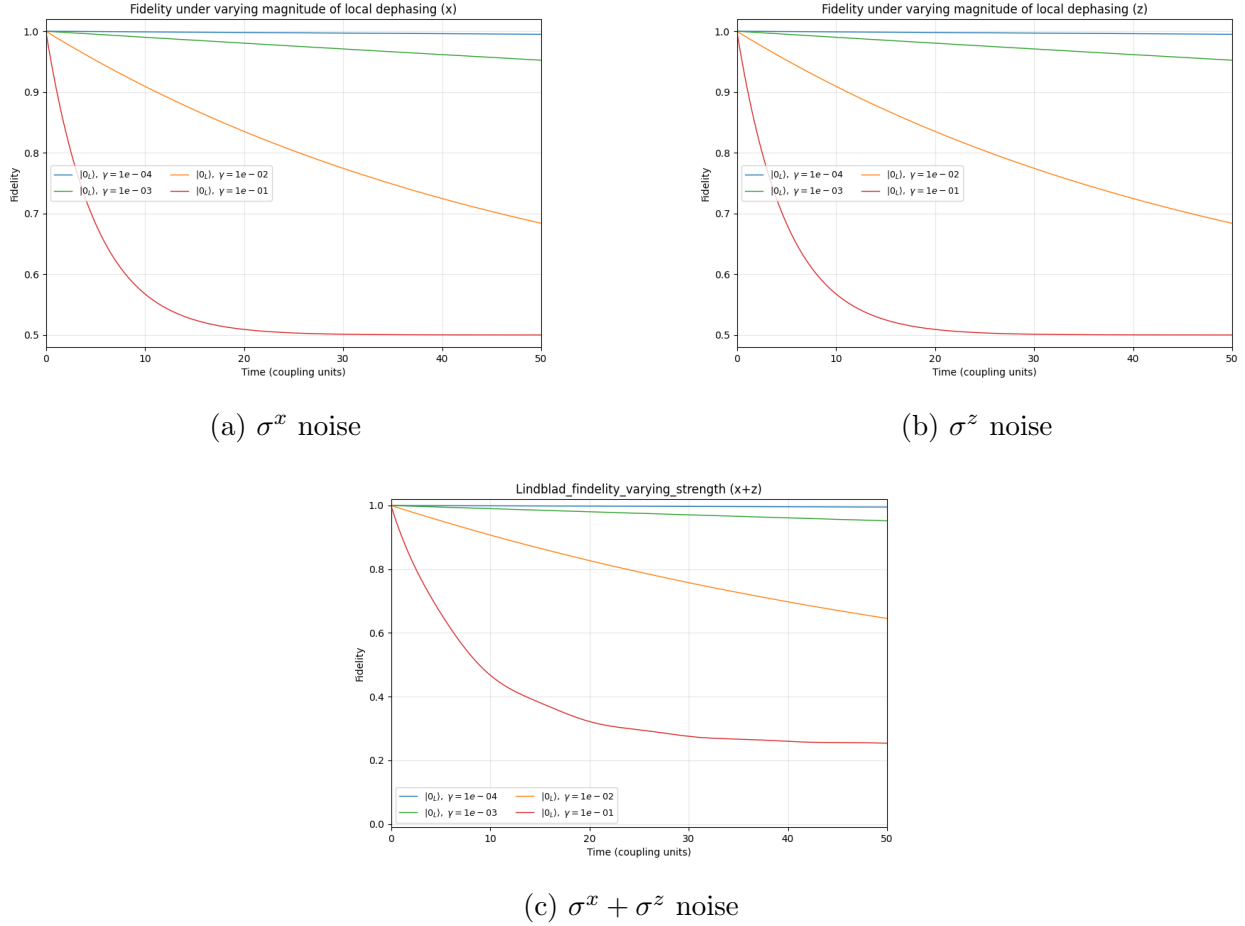


Figure 5: Effects of various noise types (σ^x , σ^z or both) in master equation evolution.

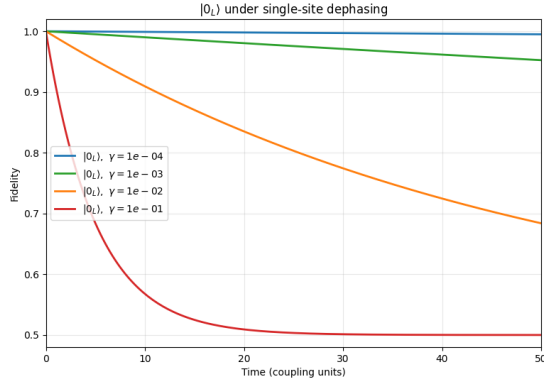
parameters considered here, the σ^x and σ^z noise channels produce very similar fidelity decay, but the superposition case does tend to be more perturbative as fidelity drops more quickly. All the coupling units on the horizontal axes are in natural units of $\hbar/t$. Another interesting investigation is to consider changing the eigenstates (or their superposition) while keeping the same simple $\sqrt{\gamma}\sigma^z$ noise. The results are found in *Figure 6*, and one can confirm the expected exponential decay and exponentially-decaying oscillations discussed above. It is crucial to note that there is no fundamental difference between $|0\rangle$ and $|1\rangle$ here, so both are valid symmetry-protected states for logical qubits.

Additionally, one could also evaluate the population in the symmetry-protected logical subspace, defined as

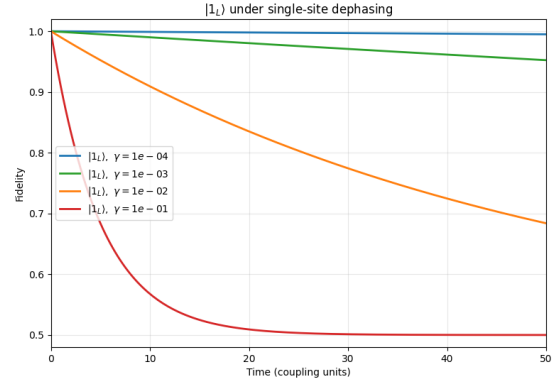
$$P_L(t) = |\langle 0_L | \rho(t) | 0_L \rangle| + |\langle 1_L | \rho(t) | 1_L \rangle|, \quad (10)$$

which would follow almost exactly the same shape as for *Figure 5*, since the effect of decoherence is essentially equivalent to decoupling from the protected regimes into unprotected subspaces.

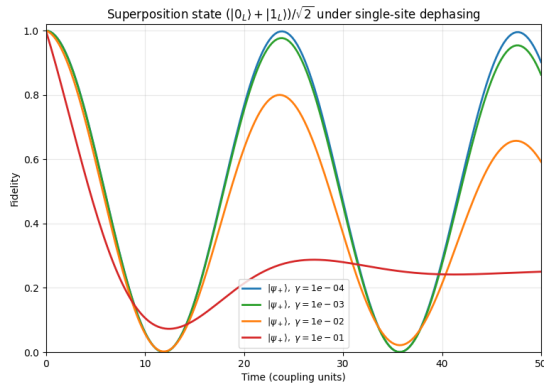
Generally, we observe that the decaying exponential envelope for fidelity has a characteristic decoherence time τ following the rough order of magnitude $\tau \sim 1/\gamma$, and certainly a higher



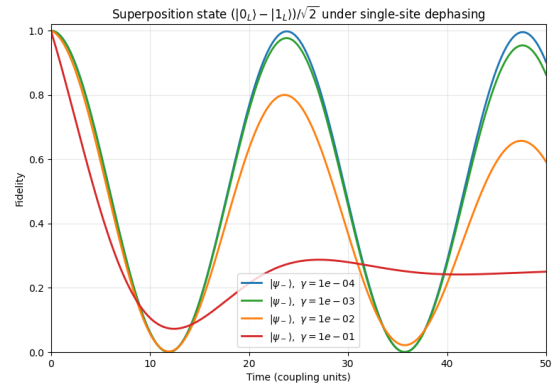
(a) $|0\rangle$



(b) $|1\rangle$



(c) $1/\sqrt{2}(|0\rangle + |1\rangle)$



(d) $1/\sqrt{2}(|0\rangle - |1\rangle)$

Figure 6: Effects of σ^z noise at eigenstates and superposition states in master equation evolution.

magnitude of environmental coupling γ shortens the decoherence time τ .

Conclusion

In conclusion, this project has investigated the robustness of symmetry-protected quantum states in a multi-qubit spin Hamiltonian, demonstrating how appropriate nearest- and next-nearest-neighbour couplings can define a protected logical subspace in which relaxation and dephasing from local perturbations are strongly suppressed. Numerical analysis identified a symmetry-protected regime in the coupling ratio λ/t , and simulations with additional Hamiltonian terms and Gaussian disorder showed that regions of low decoherence sensitivity persist for systems with different numbers of spins. Then, the open-system analysis using the Lindblad master equation demonstrated the expected decay of fidelity under environmental coupling, with stronger coupling producing shorter characteristic decoherence times while both protected eigenstates remain suitable candidates for logical qubits. Overall, these results illustrate the potential of symmetry protection as a passive mechanism for preserving quantum information, and could motivate further investigation

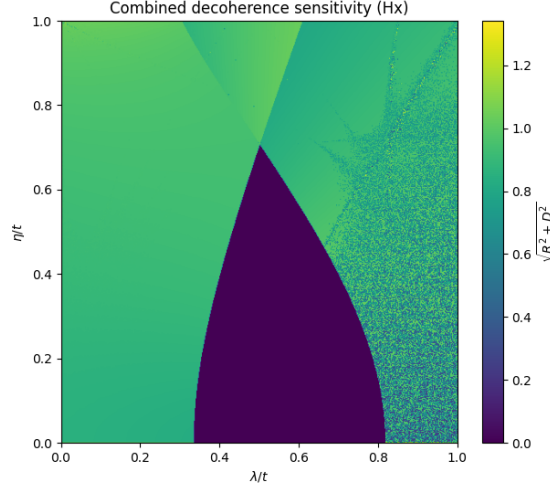
of more realistic noise channels, larger spin systems for physical implementation.

References

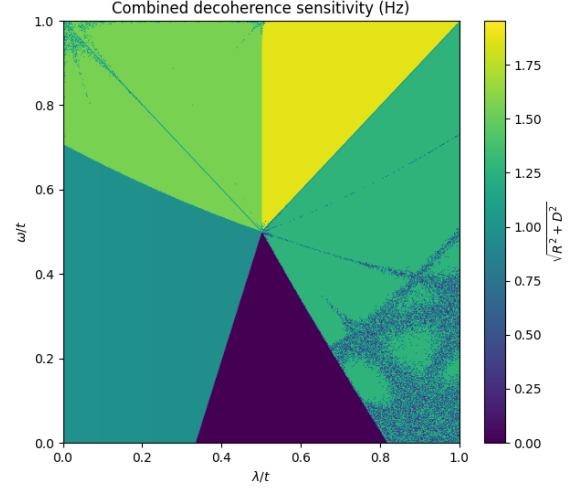
- [1] Yi Shi, Eran Ginossar, Michael Stern, and Marzena Szymanska. Symmetry-protected states of interacting qubits in superconducting quantum circuits, 2025.
- [2] Paul Brookes, Tikai Chang, Marzena Szymanska, Eytan Grosfeld, Eran Ginossar, and Michael Stern. Protection of quantum information in a chain of josephson junctions. *Physical Review Applied*, 17(2), February 2022.
- [3] Francesco Campaioli, Jared H. Cole, and Harini Hapuarachchi. Quantum master equations: Tips and tricks for quantum optics, quantum computing, and beyond. *PRX Quantum*, 5(2), 2024.

Appendix

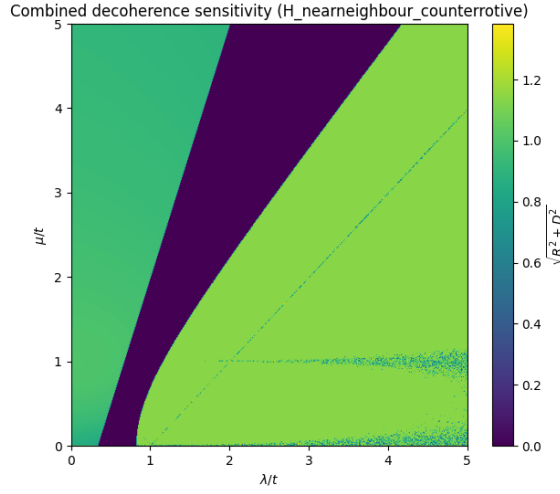
Clear views of *Figure 3* are following from next page.



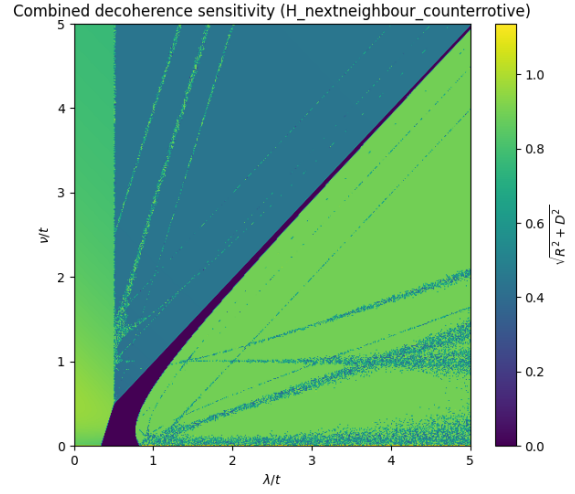
(a) $\eta/2 \sum_{n=1}^N \sigma_n^x$



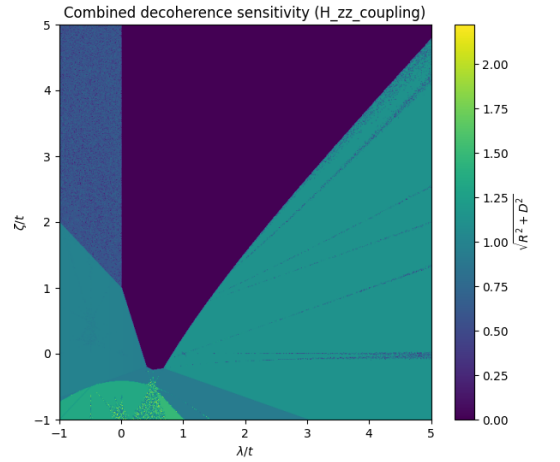
(b) $\omega/2 \sum_{n=1}^N \sigma_n^z$



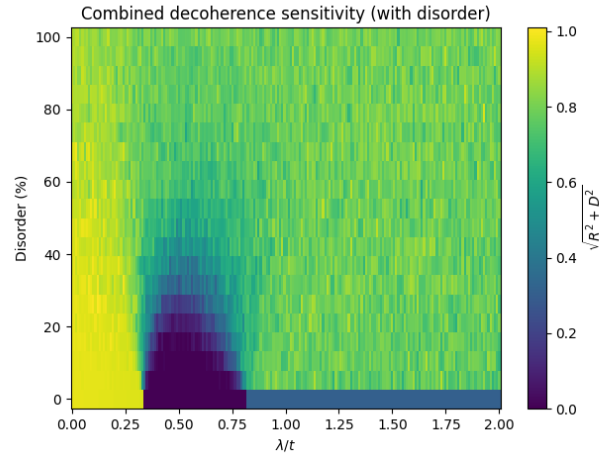
(c) $\mu/2 \sum_{n=1}^N (\sigma_n^+ \sigma_{n+1}^+ + \sigma_n^- \sigma_{n+1}^-)$



(d) $\nu/2 \sum_{n=1}^N (\sigma_n^+ \sigma_{n+2}^+ + \sigma_n^- \sigma_{n+2}^-)$

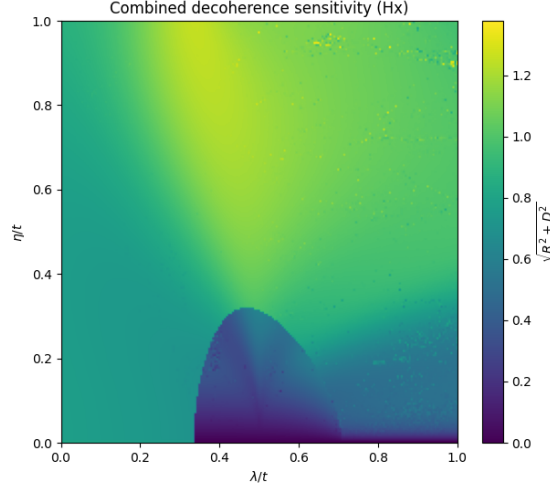


(e) $\zeta/4 \sum_{m,n=1}^N (\sigma_m^z \sigma_n^z)$

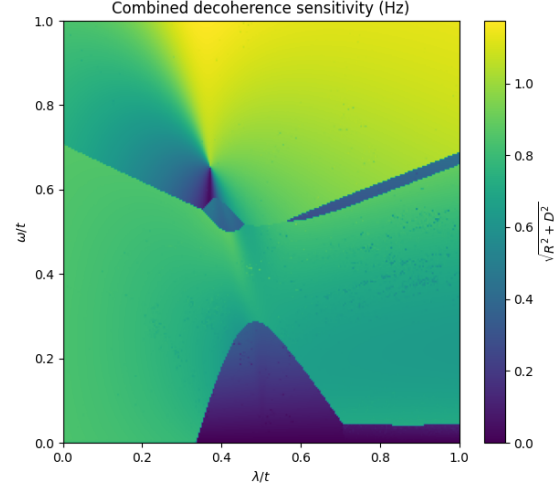


(f) Gaussian disorder (%)

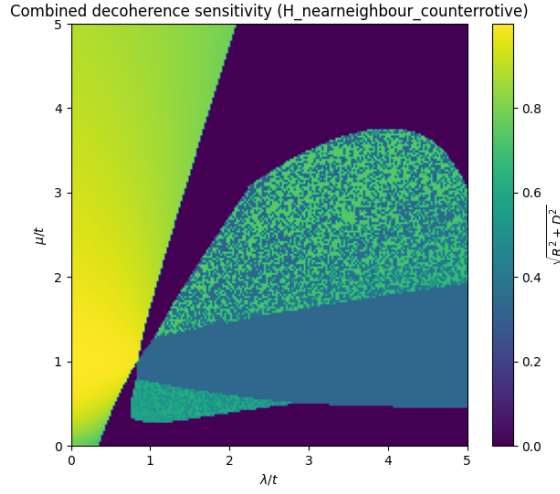
Figure 7: $N = 4$ heatmap series with various perturbation terms.



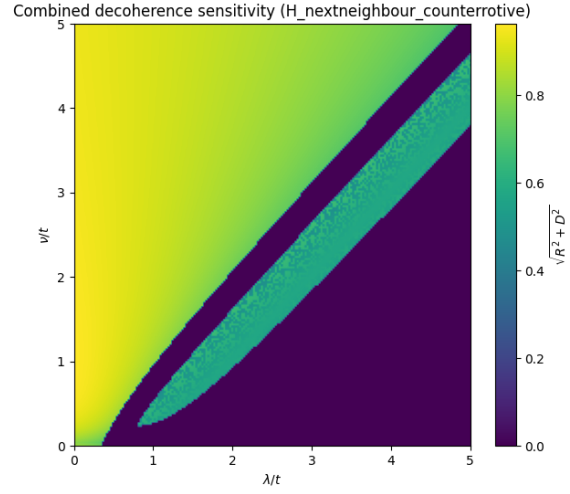
(a) $\eta/2 \sum_{n=1}^N \sigma_n^x$



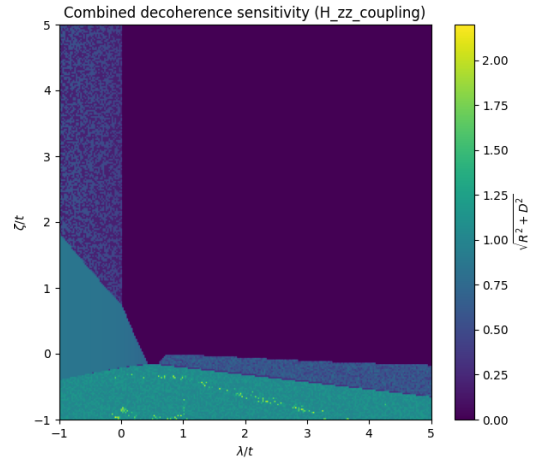
(b) $\omega/2 \sum_{n=1}^N \sigma_n^z$



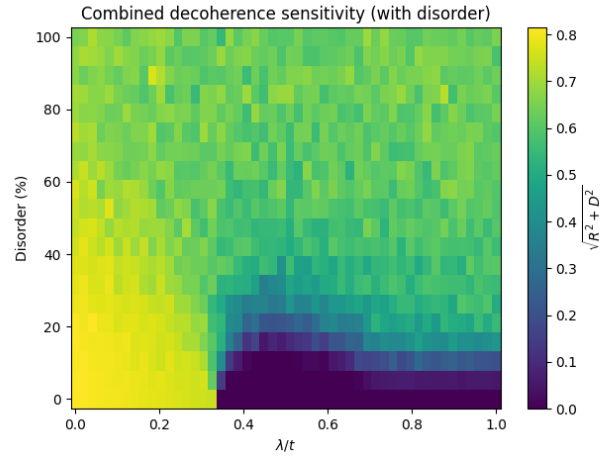
(c) $\mu/2 \sum_{n=1}^N (\sigma_n^+ \sigma_{n+1}^+ + \sigma_n^- \sigma_{n+1}^-)$



(d) $\nu/2 \sum_{n=1}^N (\sigma_n^+ \sigma_{n+2}^+ + \sigma_n^- \sigma_{n+2}^-)$

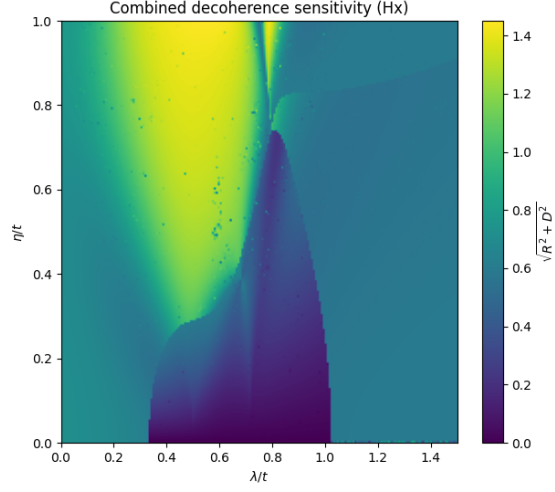


(e) $\zeta/4 \sum_{m,n=1}^N (\sigma_m^z \sigma_n^z)$

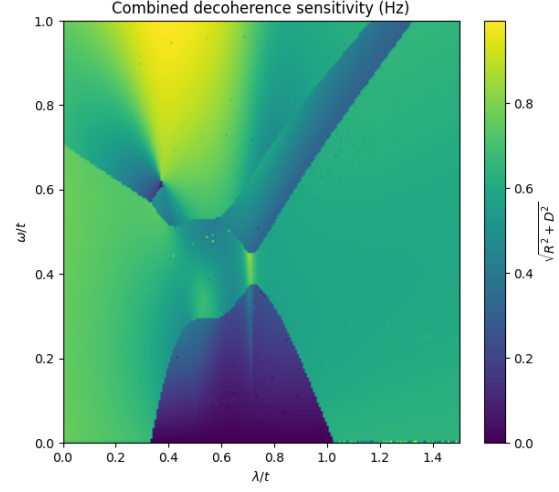


(f) Gaussian disorder (%)

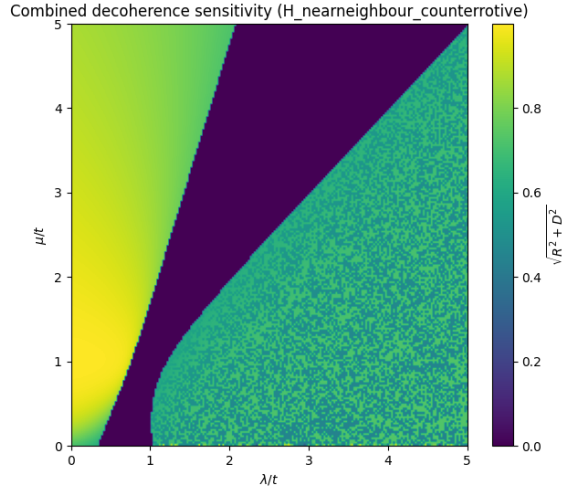
Figure 8: $N = 6$ heatmap series with various perturbation terms.



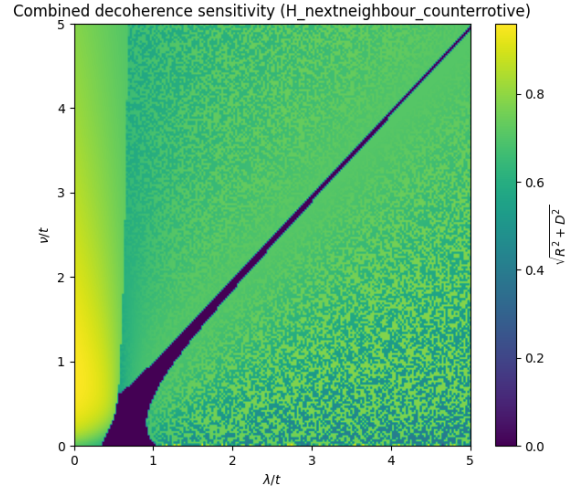
(a) $\eta/2 \sum_{n=1}^N \sigma_n^x$



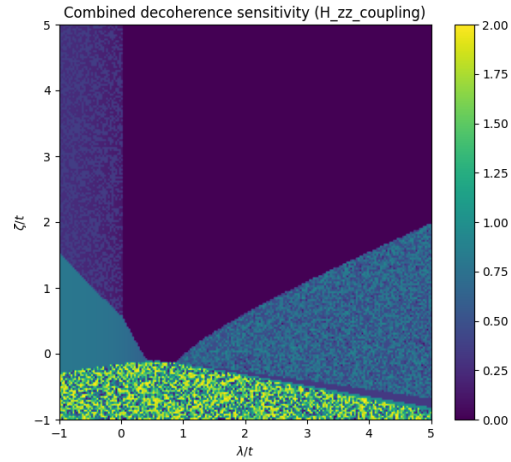
(b) $\omega/2 \sum_{n=1}^N \sigma_n^z$



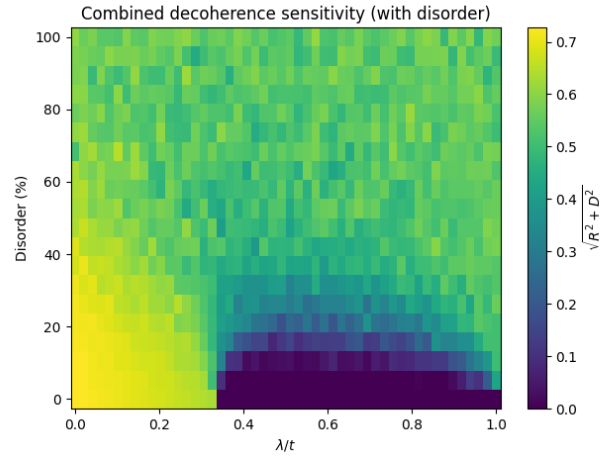
(c) $\mu/2 \sum_{n=1}^N (\sigma_n^+ \sigma_{n+1}^+ + \sigma_n^- \sigma_{n+1}^-)$



(d) $\nu/2 \sum_{n=1}^N (\sigma_n^+ \sigma_{n+2}^+ + \sigma_n^- \sigma_{n+2}^-)$



(e) $\zeta/4 \sum_{m,n=1}^N (\sigma_m^z \sigma_n^z)$



(f) Gaussian disorder (%)

Figure 9: $N = 8$ heatmap series with various perturbation terms.